Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/24/2006
PI Name:	Collicott, Steven Ph.D.		
Project Title:	Three-dimensional Capillary Interface Topologies and Stability		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	47907 - 2015	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
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No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Performance Goal No.:			
Performance Goal Text:			
	NASA-unique capillary fluid phenomena are common in the processes and hardware of human exploration of space. Recent validation of a numerical model for capillary stability and critical contact angle has opened up numerous capillary fluids science problems to scientific study for the first time. The PI is uniquely suited to perform and validate these studies. Fluids systems aboard inhabited spacecraft such as liquid fuel and cryo tanks, thermal control systems and power cycles, life support systems requiring the processing of water, and routine fluids handling operations required during microgravity experimentation and health monitoring all require these advances in fluid physics to enable more intelligent, safer, robust, and minimum-mass design. In each two-phase fluid system, capillary interfaces arise that, if not properly anticipated, can become stable or unstable when not desired, or can oscillate in resonance with other systems leading to undesirable and potentially life threatening consequences. Several potentially critical yet uninvestigated capillary phenomena arise in the fluid systems required for the human		

Task Description:	exploration of space. Such phenomena include multi-diameter tee fittings, bends, tubing joints with potential pinning ridges for contact lines, and similar in weightlessness and in low-acceleration flights as proposed for science missions. Already 3-D non-symmetric solutions are known to be the minimum energy states in special axisymmetric containers. Practical spaceflight capillary systems with some measure of symmetry are recently shown to posses similar behavior. Terrestrial applications include biological topics such as water in lung passages and various liquids in MEMS and nano-devices. Undoubtedly, any investigation along these lines of the proposed research will discover new fluid phenomena unique to the low-g environment and produce a knowledge base for use in systems design for future inhabited or robotic spacecraft. It is proposed to conduct such fundamental research while also using the science to explore applications to common and advanced spaceflight two-phase fluids control problems in support of NASA's exploration goals in space. A quantitative understanding of the stability limits and relative energies of possible topologies of capillary surfaces in clean and applied geometries will advance the state of fluid sciences. It is essential to avoid or control surface stability by design in NASA-unique applications, or to prepare for inadvertent surface stabilization or destabilization should it occur. Such an understanding is not only important to expanding the knowledge of basic fluid sciences beyond the forms that are amenable to classical analysis, but also to the successful design of spacecraft liquids systems and zero-gravity experiments and scaled models for ground-based low-g environments such as that of NASA's drop towers and low-g aircraft. Presently, the stability limits of capillary surfaces are known quantitatively only for an extremely limited set of simplified container and conduit geometries.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2005.
Bibliography Type:	Description: (Last Updated:)